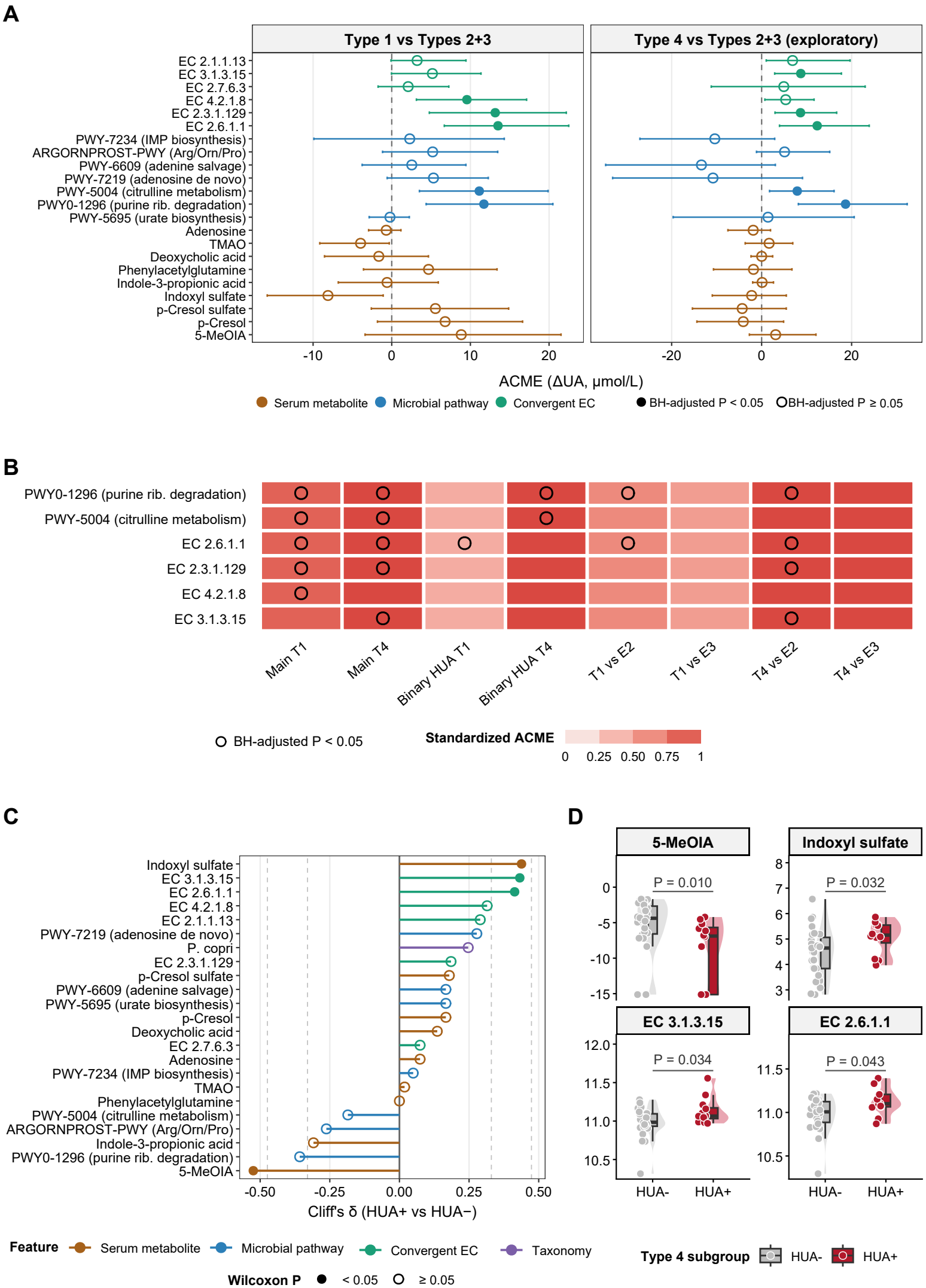

A
Type 1 vs Types 2+3
Type 4 vs Types 2+3 (exploratory)
EC 2.1.1.13
EC 3.1.3.15
EC 2.7.6.3
EC 4.2.1.8
EC 2.3.1.129
EC 2.6.1.1
PWY-7234 (IMP biosynthesis)
ARGORNPROST-PWY (Arg/Orn/Pro)
PWY-6609 (adenine salvage)
PWY-7219 (adenosine de novo)
PWY-5004 (citrulline metabolism)
PWY0-1296 (purine rib. degradation)
PWY-5695 (urate biosynthesis)
Adenosine
TMAO
Deoxycholic acid
Phenylacetylglutamine
Indole-3-propionic acid
Indoxyl sulfate
p-Cresol sulfate
p-Cresol
5-MeOIA
-10
0
10
20
-20
0
20
ACME (ΔUA, μmol/L)
Serum metabolite
Microbial pathway
Convergent EC
BH-adjusted P < 0.05
BH-adjusted P ≥ 0.05
B
PWY0-1296 (purine rib. degradation)
PWY-5004 (citrulline metabolism)
EC 2.6.1.1
EC 2.3.1.129
EC 4.2.1.8
EC 3.1.3.15
Main T1
Main T4
Binary HUA T1
Binary HUA T4
T1 vs E2
T1 vs E3
T4 vs E2
T4 vs E3
BH-adjusted P < 0.05
Standardized ACME
0
0.25
0.50
0.75
1
C
Indoxyl sulfate
EC 3.1.3.15
EC 2.6.1.1
EC 4.2.1.8
EC 2.1.1.13
PWY-7219 (adenosine de novo)
P. copri
EC 2.3.1.129
p-Cresol sulfate
PWY-6609 (adenine salvage)
PWY-5695 (urate biosynthesis)
p-Cresol
Deoxycholic acid
EC 2.7.6.3
Adenosine
PWY-7234 (IMP biosynthesis)
TMAO
Phenylacetylglutamine
PWY-5004 (citrulline metabolism)
ARGORNPROST-PWY (Arg/Orn/Pro)
Indole-3-propionic acid
PWY0-1296 (purine rib. degradation)
5-MeOIA
-0.50
-0.25
0.00
0.25
0.50
Cliff's δ (HUA+ vs HUA−)
Feature
Serum metabolite
Microbial pathway
Convergent EC
Taxonomy
Wilcoxon P
< 0.05
≥ 0.05
D
5-MeOIA
P = 0.010
Indoxyl sulfate
P = 0.032
EC 3.1.3.15
P = 0.034
EC 2.6.1.1
P = 0.043
HUA-
HUA+
Type 4 subgroup
HUA-
HUA+